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Matthew R Whiteway

ACADEMIC POSITIONS

Columbia University, New York, NY

Associate Research Scientist 2022-present
Zuckerman Mind Brain Behavior Institute

Postdoctoral Research Scientist 2018-2022
Zuckerman Mind Brain Behavior Institute
Advisor: Dr. Liam Paninski

International Brain Laboratory

Data Scientist 2022-present
Postdoctoral Research Scientist 2019-2022

University of Maryland, College Park, MD

PhD in Applied Mathematics 2012-2018
Dissertation: *A latent variable modeling framework for analyzing neural population activity*
Advisor: Dr. Daniel Butts

Undergraduate research in network science 2010
Advisors: Drs. Michelle Girvan and Ed Ott

University of Oklahoma, Norman, OK

B.Sc. in Physics, B.A. in Mathematics 2006-2011

JOURNAL

PUBLICATIONS

[A brain-wide map of neural activity during complex behaviour](#)

International Brain Laboratory, Angelaki D, Benson B, Benson J, Birman D, Bonacchi N, Bougrova K, Bruijns SA, Carandini M, Catarino JA, Chapuis GA, Churchland AK, Dan Y, Davatolhagh F, Dayan P, DeWitt EEJ, Engel TA, Fabbri M, Faulkner M, Fiete IR, Findling C, Freitas-Silva L, Gerçek B, Harris KD, Hausser M, Hofer SB, Hu F, Hubert F, Huntenburg JM, Khanal A, Krasniak C, Langdon C, Langfield C, Lau PYP, Mainen ZF, Meijer GT, Miska NJ, Mrsic-Flogel TD, Noel JP, Nylund K, Pan-Vazquez A, Paninski L, Pouget A, Rossant C, Roth N, Schaeffer R, Schartner M, Shi Y, Socha KZ, Steinmetz NA, Svoboda K, Urai AE, Wells MJ, West SJ, **Whiteway MR**, Winter O & Witten IB
Nature 2025

[Brain-wide representations of prior information in mouse decision-making](#)

Findling C*, Hubert F*, International Brain Laboratory, Acerbi L, Benson B, Benson J, Birman D, Bonacchi N, Buchanan EK, Bruijns S, Carandini M, Catarino JA, Chapuis GA, Churchland AK, Dan Y, Davatolhagh F, DeWitt EEJ, Engel TA, Fabbri M, Faulkner M, Fiete IR, Freitas-Silva L, Gerçek B, Harris KD, Hausser M, Hofer SB, Hu F, Huntenburg JM, Khanal A, Krasniak C, Langdon C, Langfield CA, Latham PE, Lau PYP, Mainen Z, Meijer GT, Miska NJ, Mrsic-Flogel TD, Noel JP, Nylund K, Pan-Vazquez A, Paninski L, Pillow J, Rossant C, Roth N, Schaeffer R, Schartner M, Shi Y, Socha KZ, Steinmetz NA, Svoboda K, Tessereau C, Urai AE, Wells MJ, West SJ, **Whiteway MR**,

- Winter O, Witten IB, Zador T, Zhang Y, Dayan P & Pouget A
Nature 2025
- [Reproducibility of *in-vivo* electrophysiological measurements in mice](#)
International Brain Laboratory, Banga K, Benson J, Bhagat J, Biderman D, Birman D, Bonacchi N, Bruijns S, Buchanan K, Campbell R, Carandini M, Chapuis GA, Churchland AK, Davatolhagh MF, Lee HD, Faulkner M, Gerçek B, Hu F, Huntenburg J, Hurwitz C, Khanal A, Krasnaik C, Langfield C, Lau P, Mackenzie N, Meijer GT, Miska NJ, Mohammadi Z, Noel JP, Paninski L, Pan-Vazquez A, Rossant C, Roth N, Schartner M, Socha K, Steinmetz NA, Svoboda K, Taheri M, Urai AE, Wang S, Wells M, West SJ, **Whiteway MR**, Winter O, Witten IB & Zhang Y
eLife 2025
- [Coordination and persistence of aggressive visual communication in Siamese fighting fish](#)
Everett CP, Norovich AL, Burke JE, **Whiteway MR**, Villamayor PR, Shih PY, Zhu Y, Paninski L & Bendesky A
Cell Reports 2025
- [Behavioral modulations can alter the visual tuning of neurons in the mouse thalamocortical pathway](#)
Socha KZ, Couto J, **Whiteway MR**, Hosseinjany S, Butts DA & Bonin V
Cell Reports 2024
- [A study of animal action segmentation algorithms across supervised, unsupervised, and semi-supervised learning paradigms](#)
Blau A, Schaffer ES, Mishra N, Miska NJ, International Brain Laboratory, Paninski L & **Whiteway MR**
Neurons, Behavior, Data analysis, and Theory 2024
- [Lightning Pose: improved animal pose estimation via semi-supervised learning, Bayesian ensembling, and cloud-native open-source tools](#)
Biderman D*, **Whiteway MR***, Hurwitz C, Greenspan N, Lee RS, Vishnubhotla A, Warren R, Pedraja F, Noone D, Schartner M, Huntenburg JM, Khanal A, Meijer GT, Noel JP, Pan-Vazquez A, Socha KZ, Urai AE, International Brain Laboratory, Cunningham JP, Sawtell NB & Paninski L
Nature Methods 2024
- [The spatial and temporal structure of neural activity across the fly brain](#)
Schaffer ES, Mishra N, **Whiteway MR**, Li W, Vancura MB, Freedman J, Patel KB, Voleti V, Paninski L, Hillman EMC, Abbott LF & Axel R
Nature Communications 2023
- [Partitioning variability in animal behavioral videos using semi-supervised variational autoencoders](#)
Whiteway MR, Biderman D, Friedman Y, Dipoppa M, Buchanan EK, Wu A, Zhou J, Bonacchi N, Miska NJ, Noel JP, Rodriguez E, Schartner M, Socha K, Urai AE, Salzman CD, International Brain Laboratory, Cunningham JP & Paninski L
PLOS Computational Biology 2021
- [Characterizing the nonlinear structure of shared variability in cortical neuron populations using latent variable models](#)
Whiteway MR, Socha K, Bonin V & Butts DA
Neurons, Behavior, Data analysis, and Theory 2019
- [Parallel processing of sound dynamics across mouse auditory cortex via spatially patterned thalamic inputs and distinct areal intracortical circuits](#)

	Liu J, Whiteway MR , Sheikhattar A, Butts DA, Babadi B & Kanold PO <i>Cell Reports</i>	2019
	Revealing unobserved factors underlying cortical activity using a rectified latent variable model applied to neural population recordings Whiteway MR & Butts DA <i>Journal of Neurophysiology</i>	2017
	Local synchronization in complex networks of coupled oscillators Stout J, Whiteway MR , Ott E, Girvan M & Antonsen TM <i>Chaos</i>	2011
CONFERENCE PUBLICATIONS	Animal behavioral analysis and neural encoding with transformer-based self-supervised pretraining Wang Y, Yu H, Blau A, Zhang Y, International Brain Laboratory, Paninski L, Hurwitz C & Whiteway MR <i>International Conference on Learning Representations</i>	2026
	Deep Graph Pose: a semi-supervised deep graphical model for improved animal pose tracking Wu A, Buchanan K, Whiteway MR , Schartner M, Meijer G, Noel JP, Rodriguez E, Everett C, Norovich A, Schaffer E, Mishra N, Salzman CD, Angelaki D, Bendesky A, International Brain Laboratory, Cunningham J & Paninski L <i>Neural Information Processing Systems</i>	2020
	Recurrent switching dynamical systems models for multiple interacting neural populations Glaser JJ, Whiteway MR , Cunningham JP, Paninski L & Linderman SW <i>Neural Information Processing Systems</i>	2020
	BehaveNet: nonlinear embedding and Bayesian neural decoding of behavioral videos Batty E*, Whiteway MR* , Saxena S, Biderman B, Abe T, Musall S, Gillis W, Markowitz JE, Churchland AK, Cunningham J, Datta SR, Linderman S & Paninski L <i>Neural Information Processing Systems</i>	2019
OPINION PIECES	20 lessons in team science: Learning from the experience of the International Brain Laboratory International Brain Laboratory, Bayer HM, Birman D, Chapuis G, DeWitt EEJ, Freitas-Silva L, Langdon C, Laranjeira I, Lau P, Paninski L, Picard S, Tessereau C, Urai AE, Whiteway MR & Winter O <i>Neuron</i>	2026
	Beyond the algorithmic oracle: Rethinking machine learning in behavioral neuroscience Nemati N & Whiteway MR <i>The Transmitter</i>	2025
	The quest for interpretable models of neural population activity Whiteway MR & Butts DA <i>Current Opinion in Neurobiology</i>	2019
INVITED TALKS	<i>Computer vision foundation models for analyzing animal behavior</i> Cosyne workshop: Inferring neural latent states from behavior	March 2026
	<i>Computer vision models for analyzing animal behavior</i> Toward an AI Neuroscientist Workshop, Yale	February 2026

<i>Improving accessibility to quantitative phenotyping in animal models</i> SFARI Workshop on unbiased approaches to quantifying motor and other observable behaviors in humans with autism	January 2026
<i>Lightning Pose: improved pose estimation via semi-supervised learning and Bayesian ensembling</i> Dutch Brain Interface Initiative	June 2025
<i>Lightning Pose: improved pose estimation via semi-supervised learning and Bayesian ensembling</i> Yale Neuro-AI Journal Club	April 2025
<i>Partitioning variability in animal behavioral videos using semi-supervised variational autoencoders</i> Flatiron Institute, Center for Computational Neuroscience	February 2024
<i>Lightning Pose: improved pose estimation via semi-supervised learning and Bayesian ensembling</i> Sainsbury Wellcome Centre Seminar	January 2024
<i>Deep learning tools for behavioral video analysis</i> UCLA Seminar	July 2023
<i>Exploiting unlabeled frames to build better models for behavioral video analysis</i> With Anqi Wu, Kelly Buchanan & Liam Paninski Minisymposium on “Modern computational techniques for tracking behavior” Neuromatch 3.0	October 2020
<i>Revealing unobserved sources of variability in populations of sensory cortical neurons</i> CCEBH/NIDCD Joint Workshop, University of Maryland	October 2016

SELECTED CONFERENCE ABSTRACTS	<i>Animal behavioral analysis and neural encoding with transformer-based self-supervised pretraining</i> Wang Y, Yu H, Blau A, Zhang Y, International Brain Laboratory, Paninski L, Hurwitz C & Whiteway MR Computational and Systems Neuroscience, Lisbon, Portugal	2026
	<i>An uncertainty-aware framework for data-efficient multi-view animal pose estimation</i> (selected as Spotlight Paper) Aharon L, Lee K, Sikka K, Chettih S, Hurwitz C, Paninski L & Whiteway MR Assoc. for the Advancement of AI: Neuro for AI & AI for Neuro Workshop, Singapore	2026
	<i>Improved calibration-free multi-view pose estimation with the variance-inflated Ensemble Kalman Smoother</i> (selected for Oral Presentation) Aharon L*, Lee K*, Sikka K, Chettih S, Hurwitz C, Paninski L & Whiteway MR Computer Vision and Pattern Recognition: CV4Animals Workshop, Nashville, TN	2025
	<i>Semi-supervised animal action segmentation with switching nonlinear dynamical systems</i> Blau A, Mishra N, Schaffer ES, Paninski L & Whiteway MR Computational and Systems Neuroscience, Lisbon, Portugal	2024
	<i>Pose estimation made better, easier, and faster with video semi-supervised learning on the cloud</i> Biderman D*, Whiteway MR* , Hurwitz C, Greenspan N, Lee R, Vishnubhotla A, Schartner M, Huntenburg J, Warren R, Noone D, Pedraja F,	

International Brain Laboratory, Sawtell N & Paninski L Computational and Systems Neuroscience, Montreal, Canada	2023
<i>Semi-supervised sequence modeling for improved behavioral segmentation</i> Whiteway MR , Schaffer ES, Wu A, Buchanan EK, Onder OF, Mishra N & Paninski L Computer Vision and Pattern Recognition: CV4animals Workshop (virtual)	2021
<i>BehaveNet: behavioral video embedding and neural analysis toolbox</i> Whiteway MR* , Batty E*, Saxena S, Biderman B, Abe T, Musall S, Gillis W, Markowitz JE, Churchland AK, Datta SR, Linderman S & Paninski L. Computational and Systems Neuroscience, Denver, CO	2020
<i>Decoding neural population activity within a latent variable framework</i> Whiteway MR , Bartolo R, Averbek BB & Butts DA Computational and Systems Neuroscience, Denver, CO	2018
<i>Nonlinear latent variable approaches for understanding population activity in sensory cortex</i> Whiteway MR , Socha K, Bonin V & Butts DA Computational and Systems Neuroscience, Salt Lake City, UT	2017

TEACHING
EXPERIENCE

Lecturer

Integral Calculus, University of Maryland	Summer 2014
Introductory Statistics, University of Maryland	Spring 2013

Teaching Assistant

Introductory Statistics, University of Maryland	Spring 2015
Multivariable Calculus, University of Maryland	Fall 2014
Integral Calculus, University of Maryland	Summer 2014
Multivariable Calculus, University of Maryland	Spring 2014
Linear Algebra, University of Maryland	Fall 2013
Introductory Statistics, University of Maryland	Spring 2013
Integral Calculus, University of Maryland	Fall 2012

Guest Lecturer

<i>Quantitative analysis of animal behavioral video data</i> Topics in Sensorimotor Control, Columbia University	Spring 2026
<i>Understanding generative models for large-scale datasets</i> DataSAI for Neuroscience Summer School, Caltech	Summer 2023
<i>Deep learning tools for behavioral video analysis</i> Statistical Analysis of Neural Data, Columbia University	Fall 2021-2025
<i>Deep learning tools for behavioral video analysis</i> Neuro-AI, University of Florida	Fall 2021, Spring 2023
<i>Methods for static and sequential clustering</i> Advanced Topics in Theoretical Neuroscience, Columbia University	Spring 2020
<i>Static dimensionality reduction methods</i> Advanced Topics in Theoretical Neuroscience, Columbia University	Spring 2019

Tutorials

Lightning Pose video tutorial series	Fall 2024
UCL Neuropixels Course: IBL visualization	Fall 2024

REVIEWING AND SERVICE

Workshop (Co-)Organizer

Workshop on Open-Source Tools for Behavioral Quantification Mediterranean Neuroscience Society, Chania, Crete	2025
Workshop on Interpretable Computational Neuroscience Computational and Systems Neuroscience, Denver, CO	2020

Reviewer

Neural Information Processing Systems (NeurIPS); NeurIPS Workshops;
International Conference on Machine Learning (ICML);
International Conference on Learning Representations (ICLR);
Nature Methods; Artificial Intelligence Reviews;
PLOS Computational Biology; eLife;
Neurons, Behavior, Data Analysis, and Theory (NBDT); Cosyne

2019-present

GRANTS AND FUNDING

NIH 1R50NS145433-01, PI <i>Enhancing the International Brain Laboratory's video processing pipeline with advanced pose estimation and action segmentation tools</i>	2025-2028
Center for Comparative and Evolutionary Biology of Hearing Trainee Grant	2015-2016

HONORS AND AWARDS

University of Maryland Department of Mathematics Excellence in Teaching Award	2013
University of Oklahoma Department of Physics and Astronomy Fowler Prize	2011
University of Oklahoma Department of Physics and Astronomy J. Clarence Karcher Scholarship	2009-2011
National Merit Scholarship	2006-2011